



Applied nutritional investigation

The effect of regular consumption of four low- or no-calorie sweeteners on glycemic response in healthy women: A randomized controlled trial

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ABSTRACT

Objectives: The aim of this study was to determine the effects of regular exposure to certain low- or no-calorie sweeteners (LNCS) on glucose tolerance and glucagon-like peptide 1 (GLP-1) release in healthy individuals.

Methods: It was designed as a randomized, single-blinded, controlled study. Healthy and normoglycemic adults who did not have regular consumption of LNCS were recruited. Participants underwent a 75-g oral glucose tolerance test (OGTT) at baseline and were randomly assigned to consume 330 mL water sweetened with saccharine, sucralose, or aspartame + acesulfame-K (Asp+Ace-K), or plain water for the control group, daily for 4 wk. Fasting plasma glucose, insulin, GLP-1, and glycated hemoglobin A_{1c} (HbA_{1c}) levels and 1-h, 2-h, and 3-h plasma glucose and insulin levels during OGTT were obtained at baseline. The change in insulin sensitivity was assessed by both the Homeostatic Model Assessment Insulin Resistance (HOMA-IR) Index and the Matsuda Index. Anthropometric measurements and dietary intakes were determined at baseline. Baseline measurements were repeated at week 4.

Results: Of the participants enrolled in the study, 42 (age, 21.24 ± 2.26 y; body mass index, 20.65 ± 2.88 kg/m²) completed the 4-wk intervention period. There were no differences for glucose, insulin, GLP-1, or HbA_{1c} levels or HOMA-IR scores at baseline or at week 4 when compared with the control group. The area under the curve of mean glucose and insulin values during OGTT were also found to be similar between groups at baseline and week 4. There were also no effects of LNCS intake on body weight, body composition, and waist circumference.

Conclusions: These results suggest that regular consumption of LNCS-sweetened water similar to doses consumed in daily life over 4 wk had no significant effect on glycemic response, insulin sensitivity, GLP-1 release, and body weight in healthy individuals. This trial was registered at www.clinicaltrials.gov as NCT04904133.

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The authors' responsibilities were as follows: S. E. O., G. S., and M.B. designed the study; S. E. O. conducted the study and had primary responsibility for the final content; G. S. oversaw the OGTT and incretin hormone analysis; M. B. oversaw the dietary data and statistical analysis; and all authors read and approved the final manuscript.

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Introduction

Low- or no-calorie sweeteners (LNCS) provide sweet taste, although they usually do not provide energy or alter blood glucose levels [1]. Therefore, they have gained increased attention by individuals who aim for weight loss or maintenance by reducing energy intake or who want to achieve glycemic control [2,3].

Low- or no-calorie sweeteners are generally recognized as a safe and healthy alternative to caloric sweeteners for weight management and glucose tolerance [1]. Safety of LNCS is determined by several national or international food safety authorities [4–6]. Furthermore, several professional societies hold the view that when used within the specified intake levels (acceptable daily intake [ADI]), they could help reductions in energy intake, thereby

resulting in weight loss or control and beneficial effects on related metabolic parameters [1,7–9]. However, some *in vitro* [10–12] and animal trials [13] have led to increased concerns about the long-term metabolic effects of LNCS consumption.

Despite their wide use and long-term safety outcomes, human studies on chronic exposure to LNCS and metabolic outcomes are still lacking, and concerns remain [13,14]. Results from meta-analyses of randomized controlled trials (RCTs) conclude that there is a beneficial or neutral effect of LNCS consumption on body weight compared with sugar-sweetened or water alternatives [15–18]. However, some meta-analyses of observational studies reported a link between consumption of LNCS and body weight gain, obesity, and some adverse metabolic outcomes [19–21]. One explanation for this inverse relationship may be the interaction of LNCS with the sweet-taste receptors (STRs) [22].

In the oral cavity, sweet compounds bind to STRs known as type 1, member 2 (T1R2) and type 1, member 3 (T1R3), and the activation of STRs generate the perception of a sweet taste in the brain [23]. Despite their differences in chemical structure, all sweet compounds (caloric, noncaloric, and sweet proteins) bind to and activate the T1R2 and T1R3 receptors in the oral cavity [24].

In recent years, it has been discovered that STRs are found not only in the mouth but also in tissues such as the stomach, small intestine, large intestine, pancreas, and brain, and sweet compounds have been shown to bind to these receptors in these tissues [25–27]. Although the tissue-specific roles of STRs have not been fully understood, they have been found to be associated with the release of insulin and incretin in the gastrointestinal tract and pancreas, as well as a perception of sweet taste [27]. In *in vitro* and animal studies, some LNCS have also been shown to bind with these receptors, providing a perception of sweet taste and affecting the release of incretin [10, 28, 29].

Differences in the biological fate of various LNCS may result in different opportunities in binding T1R2 and T1R3 receptors and stimulation of gastrointestinal peptides [30]. For example, it has been reported that sucralose promoted the release of glucagon-like peptide 1 (GLP-1) in the L cell line [10]. On the other hand, exposure to saccharin, sucralose, and acesulfame-K *in vitro* have

been shown to elicit the release of insulin in pancreatic β cells [31]. Therefore, LNCS that are delivered completely or as digestion products to the bloodstream could interact with STRs in tissues such as the brain and pancreas [32].

In addition, LNCS may possibly affect the cephalic phase response and alter gut microbiota, consequently inducing glucose intolerance [13,33,34]. However, only a few studies have assessed the long-term effects of LNCS on glucose tolerance and release of insulin and GLP-1 in humans [35–37]. There were some limitations of those studies, and a larger, longer-term controlled study with different LNCS at doses similar to daily life in a healthy population was needed. Therefore, in this study, we aimed to investigate the effects of regular consumption of LNCS-sweetened water on glucose tolerance, release of insulin, and GLP-1 levels in healthy individuals. The primary outcome was the glycemic response and insulin sensitivity; secondary outcomes were GLP-1 concentrations, body weight, body composition, and waist circumference.

Material and methods

Design and setting

This was a 4-wk, single-center, randomized, single-blinded, and controlled study with four parallel groups. It was conducted at Acıbadem Dr. Şinasi Can (Kadıköy) Hospital, Turkey.

The study protocol was approved by the Medical Research Ethics Committee (ATADEK), Acıbadem University, Istanbul, Turkey. All the participants provided signed informed consent before enrollment. This study was registered at www.clinicaltrials.gov as NCT04904133.

Participant selection

A total of 59 participants responded to the announcement for the study, and 48 of them were randomly assigned to the study groups between April 2, 2019, and May 12, 2019. A flow diagram of the participants according to the CONSORT guidelines is shown in Figure 1. The number of participants who completed the study in the saccharine, sucralose, aspartame + acesulfame-K (Asp+Ace-K), and control groups were 11, 11, 11, and 9, respectively.

Inclusion criteria were as follows: healthy, female, aged 19–45 y, body mass index (BMI) of 18–25 kg/m², not taking any medication that may affect glucose metabolism and appetite, having a regular menstrual period, having a fasting blood glucose between 70 and 100 mg/dL via capillary finger-stick, and being a

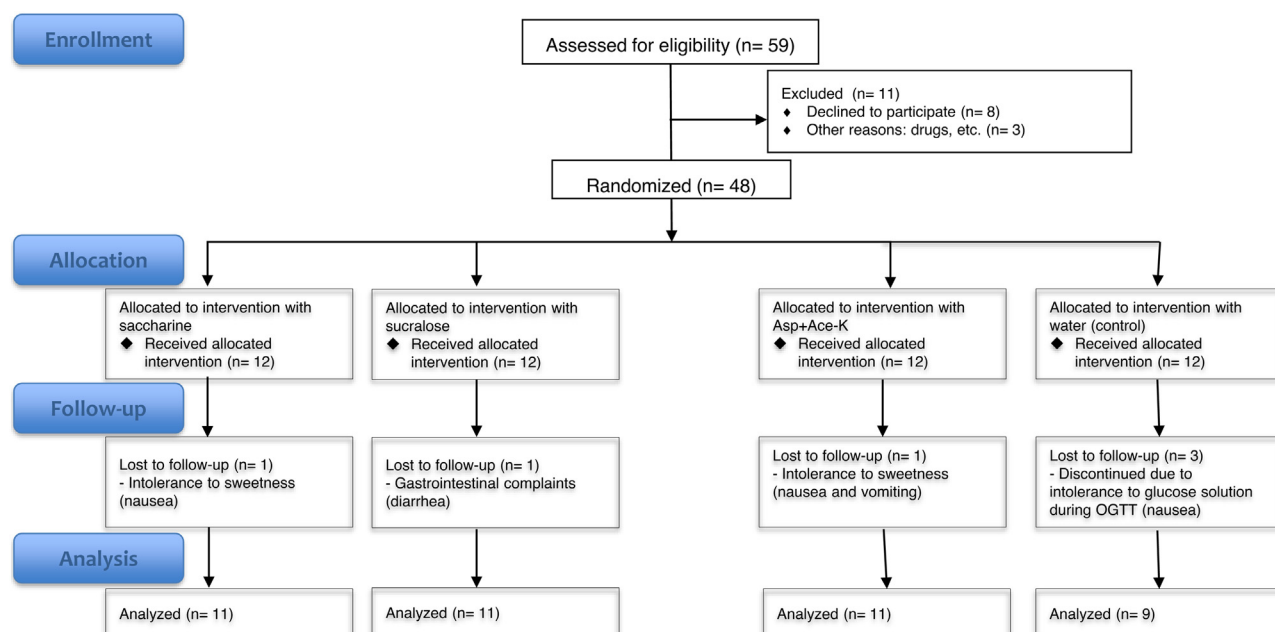


Fig. 1. Flow chart of the study.

non-consumer of LNCS (based on self-report of participants as low [<1 time/wk] or no LNCS use during the prescreening assessment).

Exclusion criteria were insulin resistance, prediabetes, diabetes (prediagnosed by a doctor), use of medication that may affect glucose metabolism (thiazide diuretics, glucocorticoids, estrogen or beta blockers), presence of acute or chronic infection, chronic alcohol intake, and regular consumption of diet soda (more than one can of soda per wk). Men and women differ in regard to their body composition, fat distribution, energy balance, sex hormones, and adipokines. Especially in the reproductive period of life, women have differences in energy partitioning that favor energy storage in subcutaneous adipose tissues and preserve them from visceral fat accumulation. In addition, women have higher insulin sensitivity than men. These differences may play a role in insulin and glucose homeostasis [38,39]. Therefore, the study included only women.

Intervention

Volunteers were randomly and equally divided into four groups (saccharin, sucralose, Asp+Ace-K, and control) using a simple randomization method (Excel 2010, Microsoft) by a staff member outside the research team and were asked to drink the LNCS-sweetened water once a day for 4 wk. The assignment of this intervention was blinded for participants. One bottle of LNCS-sweetened water (330 mL) was provided for participants daily for the weekdays, and two bottles of drink were provided on Fridays for the weekends. All bottles including those provided to the control group had the same appearance. Participants were asked to start drinking at the beginning of a meal and also were reminded daily not to forget their drink via an online message group. The concentrations of LNCS were selected to have equal sweetness to 2 cans of commercial soda. Study groups and the concentrations of LNCS they consumed were as follows:

- Group 1: 330 mL water + 140 mg saccharine
- Group 2: 330 mL water + 66 mg sucralose
- Group 3: 330 mL water + 88 mg aspartame and 88 mg acesulfame-K
- Group 4: 330 mL plain water

These doses were far less than the ADI levels proposed by the US Food and Drug Administration (FDA) or European Food Safety Authority (EFSA) [4,5]. Doses, calculated per kg of body weight, were found to be 3.3–1.6 mg for saccharine, 1.6–0.8 mg for sucralose, and 2.1–1.0 for Asp+Ace-K, according to the lowest (42 kg) and highest body weights (87 kg) of the participants. In all groups, intake of LNCS fell within respective ADI ranges.

Study design

Food and energy intake were obtained by 24-hour food recall three times (first day [baseline], 15th day, and week 4) during the study period. The 24-hour recalls were completed at the hospital under supervision of a dietitian. Dietary variables included intake of total energy, lipids, proteins, and carbohydrates.

The International Physical Activity Questionnaire (IPAQ)—Short Form was administered at the beginning of the study to estimate the participants' level of physical activity [40]. Saglam et al. [41] have reported the validity and reliability of the Turkish version of this questionnaire. Participants were asked to maintain a similar daily pattern of physical activity and nutritional habits during the study period.

An assessment visit was done on the first and last days of the study, which included measurements of body weight, height, and waist circumference, analysis

of body composition by bioelectrical impedance method, collection of blood samples, and an oral glucose tolerance test (OGTT).

Researchers in a standardized protocol obtained all anthropometric measurements. Body weight and composition were measured by a Tanita body composition analyzer (MC 180; Tanita, Tokyo, Japan). After an overnight fast, participants were instructed to remove their shoes, socks, and excess clothing before weighing. Height was measured at baseline with a calibrated stadiometer (Seca, Hamburg, Germany). To measure height, participants stood on the stadiometer with bare feet and their heads were positioned horizontally. Waist circumference was measured following a normal expiration with the arms relaxed at the sides at the mid-point between the lower part of the last rib and the top of the hip.

After an overnight fast, a peripheral venous catheter was placed in an arm, and a blood sample was collected for assessment of fasting glucose, glycated hemoglobin (HbA_{1c}), insulin, and GLP-1 levels. Next, participants ingested a 250-mL, 75-g glucose solution within 5 min, and blood samples were collected at 60, 120, and 180 min to assess blood glucose and insulin concentrations. Blood samples for GLP-1 measurement were centrifuged, and the plasma was stored at -80°C for subsequent analysis. All samples for a participant were measured on the same enzyme-linked immunosorbent assay plate. Insulin sensitivity was assessed by both the Matsuda Index and the Homeostatic Model Assessment Insulin Resistance (HOMA-IR) formula. The Matsuda Index represents whole-body insulin sensitivity, and in this study, it was calculated from values obtained at 0, 60, and 120 min during OGTT [42]. The HOMA-IR value was calculated from fasting glucose and insulin concentrations according to the following formula:

$$\text{HOMA-IR} = (\text{fasting glucose [mg/dL]} \times \text{fasting insulin [uIU/mL]}) / 405 \text{ [43]}.$$

Statistical analysis

Only data from the 42 participants who completed the 4-wk study period were included in the statistical analysis. Data were analyzed with NCSS 2007 (Kaysville, UT, USA). *P* values <0.05 were considered to be significant.

Data were reported as means and standard deviations. The Shapiro-Wilk test was used to assess normality of each variable. Differences between baseline and week 4 of each group were evaluated by the Wilcoxon signed rank test. One-way analysis of variance or the Kruskal-Wallis test was used for analysis of differences between all groups at the same time point (baseline or week 4). To adjust multiple comparisons, the Bonferroni-Dunn test was used. The Friedman test was also used to compare food intake data. Areas under the curve (AUCs) from OGTT were also calculated for each group by the trapezoid method.

The *a priori* sample size was calculated using the G*Power 3.1.9.6 program with an estimated effect size of 0.50, which is consistent with the AUC measurements of mean glucose/OGTT curves by Bueno-Hernandez et al. [44]. A total of 48 participants were needed to ensure a power of 0.80 and an α error of 0.05 for four independent groups.

Results

Baseline characteristics

Baseline characteristics are shown in Table 1. All participants had normal OGTT results at baseline. There were no significant group differences for baseline age, anthropometric measures, biochemical measures, and IPAQ-Short Form scores.

Table 1
Baseline characteristics*

	Saccharine	Sucralose	Asp+Ace-K	Control	<i>P</i> value
Participants, No.	11	11	11	9	—
Age, y	21.18 $\pm$ 1.40	21.82 $\pm$ 3.16	21.64 $\pm$ 2.54	20.11 $\pm$ 1.05	0.26
Weight, kg	59.07 $\pm$ 14.06	55.62 $\pm$ 8.52	57.35 $\pm$ 7.51	59.12 $\pm$ 8.12	0.81
Height, cm	166.82 $\pm$ 6.11	163.73 $\pm$ 4.56	167.36 $\pm$ 3.91	170.56 $\pm$ 3.91	0.024 [†]
BMI, kg/m ²	21.06 $\pm$ 3.91	20.70 $\pm$ 2.67	20.46 $\pm$ 2.43	20.3 $\pm$ 2.61	0.10
Waist circumference, cm	74.64 $\pm$ 11.19	73.18 $\pm$ 6.37	71.32 $\pm$ 5.47	72.00 $\pm$ 4.95	0.93
Fasting glucose, mg/dL	91.27 $\pm$ 6.21	91.27 $\pm$ 3.95	90.18 $\pm$ 6.08	93.89 $\pm$ 3.48	0.34
Fasting insulin, uIU/mL	8.47 $\pm$ 2.86	6.06 $\pm$ 2.30	8.59 $\pm$ 6.63	6.63 $\pm$ 2.90	0.41
HOMA-IR Index	1.92 $\pm$ 0.73	1.38 $\pm$ 0.55	1.96 $\pm$ 1.55	1.54 $\pm$ 0.71	0.44
HbA _{1c} , %	5.41 $\pm$ 0.23	5.36 $\pm$ 0.33	5.36 $\pm$ 0.31	5.41 $\pm$ 0.38	0.95
GLP-1, pg/mL	558.60 $\pm$ 91.07	630.40 $\pm$ 63.57	589.32 $\pm$ 57.07	560.63 $\pm$ 78.95	0.15
IPAQ-short form scores	1135.3 $\pm$ 953.9	1313.4 $\pm$ 791.7	1381.5 $\pm$ 1174.7	1711.7 $\pm$ 562	0.24

Asp+Ace-K, aspartame+acesulfame-K; BMI, body mass index; GLP-1, glucagonlike peptide-1; HbA_{1c}, hemoglobin A_{1c}; HOMA-IR, Homeostatic Model Assessment Insulin Resistance; IPAQ, International Physical Activity Questionnaire.

*Values are means $\pm$ SDs unless otherwise specified; the Kruskal-Wallis test was used to compare baseline characteristics between groups.

[†]Significant difference between groups ($P < 0.05$).

Glycemic response, insulin sensitivity, HbA_{1c} and GLP-1 concentration

Baseline and week-4 plasma glucose concentrations are plotted in Figure 2A and 2B, respectively. There was a significant difference in fasting glucose concentrations between groups at week 4 ($P = 0.035$); the Asp+Ace-K group had lower plasma glucose concentrations than the saccharine group, but there was no significant difference compared with the control group. At 60, 120, and 180 min, mean glucose concentrations were similar between groups. Overall, within-group changes in plasma glucose concentrations were similar compared with baseline. There were also no statistically significant differences in mean glucose AUC values during OGTT between groups at baseline and week 4 (Table 2).

Baseline and week-4 plasma insulin concentrations are plotted in Figure 2C and 2D, respectively. There was a significant difference between groups in insulin concentrations at week 4 at 60 min ($P = 0.02$) and 120 min ($P = 0.01$); the sucralose group had lower plasma insulin concentrations than the Asp+Ace-K group, but there were no significant difference compared with the control group. At 0 and 180 min, mean insulin concentrations were similar between groups. Overall, mean plasma insulin AUC values during OGTT were similar between groups at baseline. At week 4, there were statistically significant differences between the sucralose and Asp+Ace-K groups. However, when compared with the control group, there were no differences in plasma insulin AUC values (Table 3).

Changes in the HOMA-IR and Matsuda indices, HbA_{1c} levels, and GLP-1 concentrations are shown in Table 4. The change in insulin sensitivity according to Matsuda Index values from baseline

did not significantly differ between or within groups. The HOMA-IR Index values were also similar during the study period. However, HbA_{1c} scores were significantly lower at week 4 in the saccharine ($P = 0.005$), sucralose ($P = 0.005$), and control ($P = 0.019$) groups compared with baseline. On the other hand, overall HbA_{1c} scores were similar between groups at week 4. Changes in HOMA-IR scores and HbA_{1c} levels over time were also similar between groups.

Fasting GLP-1 concentrations at baseline and week 4 are also shown in Table 4. At week 4, fasting GLP-1 concentrations were similar between groups, but compared with baseline, there was a significant decline in the sucralose ($P = 0.003$) and Asp+Ace-K ($P = 0.007$) groups. However, within-group changes from baseline were similar for all groups.

Anthropometric measurements, energy, and macronutrient intakes

Anthropometric measurements and changes during the study period are shown in Table 5. Participants were not prescribed an energy-controlled diet and were asked to maintain similar patterns of physical activity and nutritional habits during the study period. The only addition to their diet was the water-plus-LNCS mixture, which contained no or a negligible amount of calories. Thus, no weight change was expected in any group. Not surprisingly, body weight and BMI remained stable during the study period; there were no significant differences between groups at week 4. Changes in body weight and BMI values were also not significant within groups. Similarly, waist circumference at week 4 and also change

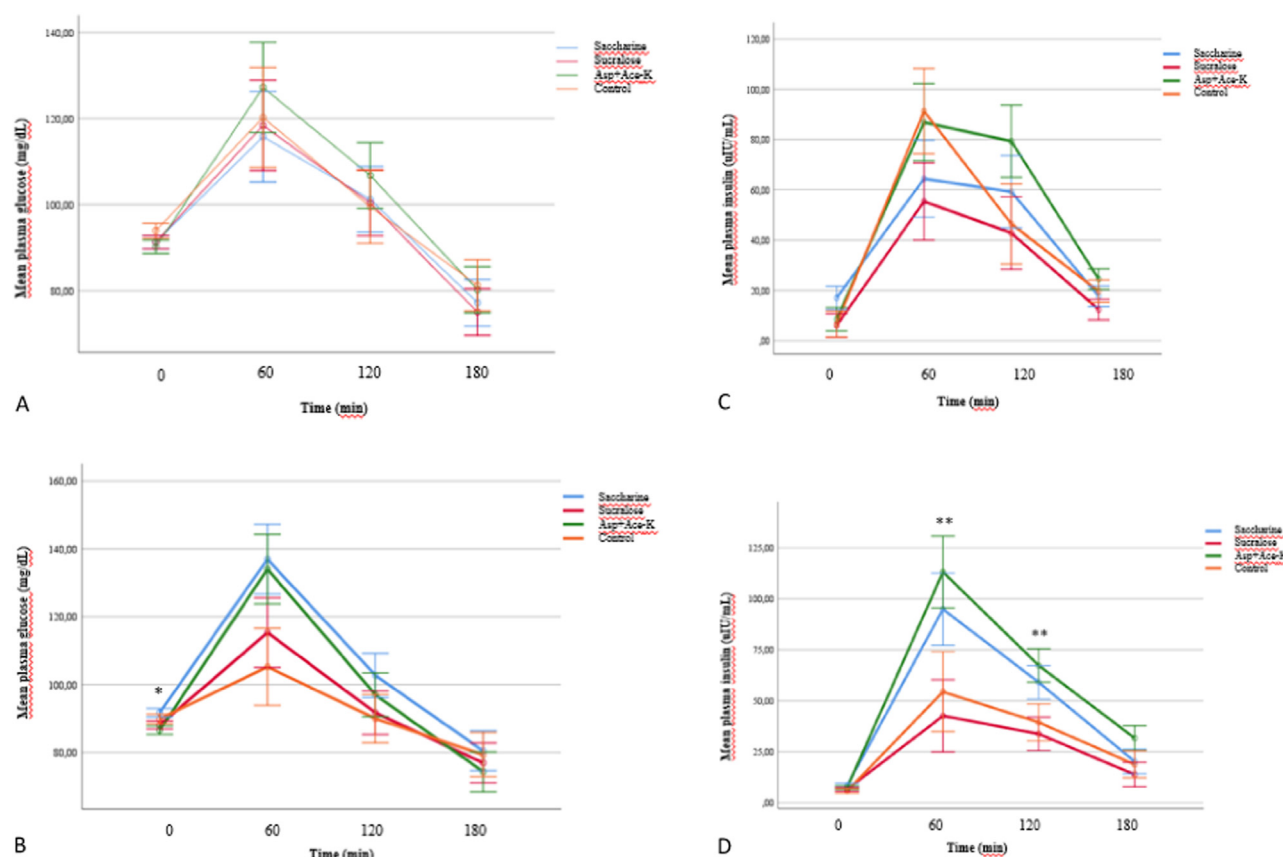


Fig. 2. Mean plasma glucose (A, B) and insulin (C, D) concentrations during the oral glucose tolerance test for participants consuming saccharine ($n = 11$), sucralose ($n = 11$), aspartame+acesulfame-K (Asp+Ace-K) ($n = 11$), or water ($n = 9$) at baseline (A, C) and at week 4 (B, D). * Difference between saccharine and Asp+Ace-K groups, $P < 0.05$. † Difference between sucralose and Asp+Ace-K groups, $P < 0.05$.

Table 2

Differences in glucose AUC values between groups at baseline and week-4 OGTT

	Mean difference	Std. error	P value *	95% Confidence interval	
				Lower bound	Upper bound
Glucose, AUC, baseline					
Saccharine					
Sucralose	−43.64	1576.72	1.00	−4432.41	4345.14
Asp+Ace-K	−1077.27	1576.72	1.00	−5466.04	3311.50
Control	−365.15	1662.01	1.00	−4991.32	4261.02
Sucralose					
Saccharine	43.64	1576.72	1.00	−4345.14	4432.41
Asp+Ace-K	−1033.64	1576.72	1.00	−5422.41	3355.14
Control	−321.52	1662.01	1.00	−4947.69	4304.66
Asp+Ace-K					
Saccharine	1077.27	1576.72	1.00	−3311.50	5466.04
Sucralose	1033.64	1576.72	1.00	−3355.14	5422.41
Control	712.12	1662.01	1.00	−3914.05	5338.29
Control					
Saccharine	365.15	1662.01	1.00	−4261.02	4991.32
Sucralose	321.52	1662.01	1.00	−4304.66	4947.69
Asp+Ace-K	−712.12	1662.01	1.00	−5338.29	3914.05
Glucose, AUC, week 4					
Saccharine					
Sucralose	2173.64	1417.49	0.80	−1771.92	6119.19
Asp+Ace-K	859.09	1417.49	1.00	−3086.46	4804.64
Control	2753.64	1494.17	0.44	−1405.34	6912.61
Sucralose					
Saccharine	−2173.64	1417.49	0.80	−6119.19	1771.92
Asp+Ace-K	−1314.55	1417.49	1.00	−5260.10	2631.01
Control	580.00	1494.17	1.00	−3578.98	4738.98
Asp+Ace-K					
Saccharine	−859.09	1417.49	1.00	−4804.64	3086.46
Sucralose	1314.55	1417.49	1.00	−2631.01	5260.10
Control	1894.55	1494.17	1.00	−2264.43	6053.52
Control					
Saccharine	−2753.64	1494.17	0.44	−6912.61	1405.34
Sucralose	−580.00	1494.17	1.00	−4738.98	3578.98
Asp+Ace-K	−1894.55	1494.17	1.00	−6053.52	2264.43

Asp+Ace-K, aspartame+acesulfame-K; AUC, area under the curve; OGTT, oral glucose tolerance test.

*One-way analysis of variance for multiple comparisons with Bonferroni.

over time was similar between and within groups. In the context of body composition (fat mass, fat-free mass, percentage of fat mass, and total body water), there was no significant difference between groups at week 4 or within groups during the study period.

Energy and macronutrient intakes during the study period are shown in Table 6. Self-reported energy and nutrient intake data from 24-hour food recalls were obtained at three different time points during study period. Reported energy intake did not differ over time in any of the groups, and baseline energy intakes were also similar between groups. Carbohydrate, protein, and lipid intakes were also similar over the study period.

Discussion

This 4-wk, single-blinded, randomized controlled study showed that consuming LNCS-sweetened water in similar doses to those consumed in daily life did not seem to affect glycemic response and GLP-1 release in healthy individuals. The glycemic response to each intervention (saccharin, sucralose, Asp+Ace-K, or control) at baseline and week 4 was measured. The data showed that responses after the ingestion of 4 different LNCS were not significantly higher than those in the control group, either at baseline or after 4 wk of LNCS consumption. Glucose, insulin, and HbA_{1c} levels and the Matsuda and HOMA-IR indices, assessed at baseline and at the end of the study, were within normal range. There were no significant differences between the LNCS and control groups in terms of changes from baseline for fasting glucose, insulin, HbA_{1c}, and GLP-1 levels and insulin sensitivity parameters. In the sucralose

and Asp+Ace-K groups, there was a significant decrease in GLP-1 levels at week 4 compared with baseline. However, no statistically significant difference was shown between groups.

These results extend the current understanding of the effects of LNCS on glycemic responses. Early interventional studies with smaller sample sizes that examined acute effects of LNCS intake showed that LNCS did not change the glycemic response in healthy individuals [45–47]. However, the discovery of STRs in the gastrointestinal tract has elicited new questions in terms of LNCS consumption and glucose tolerance and incretin release. It is hypothesized that binding of LNCS to T1R2 or T1R3 on taste and enteroendocrine cells disrupts glucose homeostasis and increases appetite [48]. In vitro studies have shown that both caloric and non-caloric sweeteners can provoke incretin (GLP-1 and Gastric inhibitory polypeptide GIP) release and increased expression of glucose transporters through their actions on intestinal STRs [10–12,49]. However, in vivo studies revealed conflicting results. Several animal studies have shown that LNCS may increase expression of glucose transporters and also glucose absorption [11,12]. On the other hand, a majority of studies failed to confirm the effects on hormone secretion [11,12,50].

Similarly, contradictory results have been reported in human studies. Previous human studies investigating short-term effects of LNCS on glycemic control indicated that there was no difference between LNCS and control groups and they had no effect on diabetes risk [47,51,52]. However, Suez et al. [13] found in their small-scale intervention study that saccharine ingestion led to elevated glycemic response in 4 of 7 participants. Recently, a meta-analysis

Table 3

Differences in insulin AUC values between groups at baseline and week-4 OGTT

	Mean difference	Std. error	P value *	95% Confidence interval	
				Lower bound	Upper bound
Insulin, AUC, baseline					
Saccharine					
Sucralose	2012.89	2381.86	1.00	−4616.94	8642.72
Asp+Ace-K	−2504.02	2381.86	1.00	−9133.85	4125.81
Control	−594.62	2510.70	1.00	−7583.08	6393.84
Sucralose					
Saccharine	−2012.89	2381.86	1.00	−8642.72	4616.94
Asp+Ace-K	−4516.91	2381.86	0.39	−11146.74	2112.92
Control	−2607.51	2510.70	1.00	−9595.97	4380.94
Asp+Ace-K					
Saccharine	2504.02	2381.86	1.00	−4125.81	9133.85
Sucralose	4516.91	2381.86	0.39	−2112.92	11146.74
Control	1909.40	2510.70	1.00	−5079.06	8897.85
Control					
Saccharine	594.62	2510.70	1.00	−6393.84	7583.08
Sucralose	2607.51	2510.70	1.00	−4380.94	9595.97
Asp+Ace-K	−1909.40	2510.70	1.00	−8897.85	5079.06
Insulin, AUC, week 4					
Saccharine					
Sucralose	4898.21	2048.71	0.13	−804.33	10600.75
Asp+Ace-K	−1905.57	2048.71	1.00	−7608.11	3796.97
Control	3724.67	2159.53	0.56	−2286.34	9735.68
Sucralose					
Saccharine	−4898.21	2048.71	0.13	−10600.75	804.33
Asp+Ace-K	−6803.78	2048.71	0.01 †	−12506.32	−1101.24
Control	−1173.54	2159.53	1.00	−7184.55	4837.47
Asp+Ace-K					
Saccharine	1905.57	2048.71	1.00	−3796.97	7608.11
Sucralose	6803.78	2048.71	0.01 †	1101.24	12506.32
Control	5630.24	2159.53	0.08	−380.76	11641.25
Control					
Saccharine	−3724.67	2159.53	0.56	−9735.68	2286.34
Sucralose	1173.54	2159.53	1.00	−4837.47	7184.55
Asp+Ace-K	−5630.24	2159.53	0.08	−11641.25	380.76

Asp+Ace-K, aspartame+acesulfame-K; AUC, area under the curve; OGTT, oral glucose tolerance test.

*One-way analysis of variance for multiple comparisons with Bonferroni.

†P < 0.05.

Table 4Changes in HOMA-IR and Matsuda indices, HbA_{1c} levels, and GLP-1 concentration

	Saccharine	Sucralose	Asp+Ace-K	Control	P value *
HOMA-IR Index					
Baseline	1.92 ± 0.73	1.38 ± 0.55	1.96 ± 1.55	1.54 ± 0.71	0.44
Week 4	1.94 ± 0.96	1.43 ± 0.26	1.6 ± 0.73	1.29 ± 0.78	0.36
P value †	0.42	0.66	0.48	0.31	—
Difference	−0.02 ± 0.77	−0.06 ± 0.67	0.36 ± 1.64	0.25 ± 0.82	0.77
Matsuda Index					
Baseline	5.88 ± 2.25	8.82 ± 4.58	7.02 ± 6.09	8.33 ± 9.12	0.65
Week 4	5.83 ± 2.96	7.84 ± 1.85	5.54 ± 3.13	9.04 ± 4.61	0.06
P value †	0.96	0.41	0.86	0.17	—
HbA _{1c} , %					
Baseline	5.41 ± 0.23	5.36 ± 0.33	5.36 ± 0.31	5.41 ± 0.38	0.95
Week 4	5.03 ± 0.23	4.95 ± 0.33	5.09 ± 0.36	5.08 ± 0.22	0.88
P value †	0.005 †	0.005 †	0.07	0.019 †	—
Difference	−0.38 ± 0.27	−0.41 ± 0.24	−0.41 ± 0.24	−0.33 ± 0.27	0.78
GLP-1, pg/mL					
Baseline	558.60 ± 91.07	630.40 ± 63.57	589.32 ± 57.07	560.63 ± 78.95	0.15
Week 4	510.05 ± 86.20	515.83 ± 104.74	496.45 ± 78.94	513.41 ± 68.86	0.94
P value †	0.53	0.003 †	0.007 †	0.33	—
Difference	−48.56 ± 138.70	−114.57 ± 116.67	−92.87 ± 74.93	−47.22 ± 106.42	0.26

Asp+Ace-K, aspartame+acesulfame-K; GLP-1, glucagonlike peptide-1; HOMA-IR, Homeostatic Model Assessment Insulin Resistance.

*Kruskal-Wallis test.

†Wilcoxon signed rank test.

‡P < 0.01.

Table 5
Anthropometric measurements

	Saccharine	Sucralose	Asp+Ace-K	Control	P value *
Body weight, kg					
Baseline	59.07 ± 14.06	55.62 ± 8.52	57.35 ± 7.51	59.12 ± 8.12	0.81
Week 4	58.87 ± 13.98	57.19 ± 7.97	57.61 ± 7.83	59.05 ± 8.09	0.95
P value †	0.21	0.27	0.35	0.81	
Difference	−0.20 ± 0.60	1.57 ± 5.21	0.25 ± 0.76	−0.07 ± 0.58	0.36
BMI, kg/m ²					
Baseline	21.06 ± 3.91	20.70 ± 2.67	20.46 ± 2.43	20.3 ± 2.61	0.10
Week 4	21.00 ± 3.92	21.28 ± 2.42	20.55 ± 2.58	20.28 ± 2.56	0.90
P value †	0.21	0.33	0.33	0.77	
Difference	−0.07 ± 0.22	0.58 ± 1.94	0.09 ± 0.28	−0.03 ± 0.20	0.37
Waist circumference, cm					
Baseline	74.64 ± 11.19	73.18 ± 6.37	71.32 ± 5.47	72.00 ± 4.95	0.93
Week 4	75.45 ± 11.67	73.23 ± 6.45	71.82 ± 6.05	72.17 ± 4.76	0.91
P value †	0.13	0.92	0.30	0.83	
Difference	0.82 ± 1.79	0.05 ± 1.81	0.50 ± 1.53	0.17 ± 1.77	0.70
Total fat mass, kg					
Baseline	15.47 ± 8.87	13.44 ± 5.50	13.90 ± 4.31	14.51 ± 5.44	0.99
Week 4	15.21 ± 8.88	13.10 ± 5.84	13.43 ± 5.21	14.55 ± 5.48	0.93
P value †	0.42	0.48	0.21	0.81	
Difference	−0.25 ± 0.88	−0.34 ± 0.94	−0.48 ± 1.15	0.04 ± 1.00	0.92
Total fat mass, %					
Baseline	24.38 ± 8.65	23.34 ± 6.63	23.79 ± 4.92	23.94 ± 6.44	0.99
Week 4	24.02 ± 8.63	22.62 ± 7.07	22.62 ± 6.59	24.07 ± 6.54	0.95
P value †	0.48	0.33	0.11	0.95	
Difference	−0.36 ± 1.31	−0.72 ± 1.82	−1.17 ± 2.02	0.12 ± 1.55	
Fat-free mass, kg					
Baseline	41.09 ± 5.21	39.88 ± 3.22	41.23 ± 3.78	42.33 ± 3.96	0.38
Week 4	41.44 ± 4.93	40.25 ± 2.78	41.86 ± 3.55	42.23 ± 4.06	0.37
P value †	0.56	0.40	0.009 ‡	0.59	
Difference	0.35 ± 1.48	0.38 ± 1.06	0.63 ± 0.65	−0.11 ± 0.99	0.34
Total body water, kg					
Baseline	31.68 ± 4.23	30.46 ± 2.36	31.18 ± 2.90	32.21 ± 2.89	0.47
Week 4	31.28 ± 3.76	30.62 ± 2.06	31.66 ± 2.73	32.14 ± 2.97	0.34
P value †	0.86	0.68	0.007 ‡	0.59	
Difference	−0.06 ± 0.80	0.16 ± 0.67	0.48 ± 0.46	−0.07 ± 0.73	0.16

Asp+Ace-K, aspartame+acesulfame-K; BMI, body mass index.

*Kruskal-Wallis test.

†Wilcoxon signed rank test.

‡P < 0.01.

Table 6
Energy and macronutrient intake

	Saccharine	Sucralose	Asp+Ace-K	Control	P value *
Total energy intake, kcal/d					
Baseline	1429.67 ± 424.30	1517.85 ± 431.89	1470.76 ± 409.78	1612.30 ± 286.90	0.64
Week 2	1418.13 ± 443.53	1577.15 ± 391.30	1542.50 ± 382.31	1490.02 ± 304.33	0.85
Week 4	1346.24 ± 406.24	1614.18 ± 373.80	1450.83 ± 439.58	1582.25 ± 248.30	0.33
P value †	0.53	0.23	0.53	0.24	
Protein intake, g/d					
Baseline	50.36 ± 18.28	52.00 ± 21.50	50.97 ± 20.36	54.93 ± 10.63	0.74
Week 2	50.09 ± 16.46	57.49 ± 18.02	53.11 ± 19.57	64.16 ± 10.31	0.25
Week 4	60.88 ± 15.38	55.97 ± 17.95	51.16 ± 17.59	55.48 ± 10.91	0.56
P value †	0.08	0.34	0.91	0.12	
Carbohydrate intake, g/d					
Baseline	142.86 ± 43.92	159.96 ± 55.95	163.00 ± 53.47	168.28 ± 48.37	0.73
Week 2	153.98 ± 56.98	167.46 ± 42.95	151.10 ± 42.92	124.24 ± 39.26	0.15
Week 4	144.02 ± 53.33	169.72 ± 41.34	146.14 ± 62.43	156.60 ± 17.59	0.68
P value †	0.91	0.53	0.44	0.10	
Lipid intake, g/d					
Baseline	71.38 ± 24.07	73.46 ± 29.91	67.24 ± 22.27	79.01 ± 23.61	0.83
Week 2	65.91 ± 24.94	72.91 ± 23.52	74.83 ± 24.44	80.24 ± 21.07	0.61
Week 4	57.15 ± 23.76	72.92 ± 26.56	71.87 ± 39.90	80.60 ± 27.06	0.20
P value †	0.31	0.70	0.76	0.90	

Asp+Ace-K, aspartame+acesulfame-K.

*Kruskal-Wallis test.

†Friedman test.

of 29 acute RCTs on humans, conducted by Nichol et al. [53], showed that intake of aspartame, saccharine, and sucralose did not have glycemic effects.

In terms of incretin secretion, most of the studies with acute intake of aspartame, sucralose, or acesulfame-K did not elicit alterations in GLP-1 release [45–47,54,55]. However, a few studies with intake of acesulfame-K and sucralose have shown increased GLP-1 secretion in healthy individuals [14,48,56] and adolescents with type 1 diabetes [57].

There are only a few studies investigating the long-term effects of LNCS on glycemic response and GLP-1 release in humans. A majority of the studies assessing the glycemic effect of aspartame [35,54,55], acesulfame-K [58], saccharine [59], and sucralose [59,60] did not show significant alterations on glucose homeostasis. However, in some studies, impaired glucose tolerance was observed [13,36,61]. In terms of incretin secretion, only three studies evaluated the effects of some LNCS; one of them using aspartame did not show any alterations on GLP-1 secretion [35]. Young et al. [36] showed attenuation in the incremental areas under the curve for GLP-1 release after exposure to a sucralose+acesulfame-K mixture. Contrary to this finding, Lertrit et al. [37] found that glucose-stimulated GLP-1 release was increased after exposure to sucralose. Differences in study participants, methods, and sample sizes may explain variable findings reported in the literature. In addition, dose, composition, route of administration, and timing of LNCS also varied between studies.

Secondly, in this study, body weight, waist circumference, body composition, and BMI values were also not affected by LNCS intake. In conjunction with this, self-reported dietary energy intakes were similar over the study period in all groups. Although some observational studies showed a positive association between LNCS intake and increased BMI [16,17,19], interventional studies with mostly RCT designs reported beneficial or neutral effects on body weight [15–17,62]. Furthermore, our study period was not long enough to expect weight changes in healthy individuals.

This study has some limitations. We assessed GLP-1 concentrations only in a fasted state. However, measuring GLP-1 along with OGTT would provide better insights. In addition, we only reminded participants to drink their beverage daily via an online message group; we did not directly measure adherence to the test drinks. In addition, the time when the LNCS was ingested daily was not recorded, but it may have had an effect on hormonal response. We used 24-hour food recall at three different time points over the study period, which could lead to inaccuracy in the estimation of dietary habits. Furthermore, our study was conducted among healthy women ages 19–45 y for a 4-wk period, and results provided in this study cannot directly be extrapolated for a different population or to a longer duration of exposure.

In conclusion, the present study showed that regular LNCS intake for 4 wk did not change glycemic responses, GLP-1 release, body weight, BMI, and body compositions in healthy individuals. A significant decrease in GLP-1 release in the sucralose and the Asp+Ace-K groups was shown compared with baseline; however, the effect was not significant in comparison with the control group. In addition, the results are valid only in the context of the 4-wk study period and for LNCS doses used in this study; they should not be directly extrapolated for longer-term use in varying doses. To our knowledge, in the literature, there is no study longer than 12 wk of duration investigating the effect of LNCS on glucose control. On the other hand, epidemiological studies suggest that LNCS may have a negative effect on glucose tolerance when used chronically for many years. In the light of the available data, LNCS

use may be a short-term substitution strategy for individuals consuming sugar-sweetened beverages regularly. However, there are no sufficient data for the effects of chronic LNCS use on metabolic health. Therefore, further studies investigating long-term effects of LNCS in different BMI groups and metabolic diseases are needed.

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